

- **e1071** includes functions for statistics and probability theory.

Scala: This language has its base in the Spark project, which supports ML through MLLIB. It has good libraries. Adoption has been low due to less community support and fewer users.

Why Python for machine learning

Python is widely being used in industry for ML applications. According to Web statistics, Python ranks second among the list of best programming languages. It is comparatively easier to pick up than other languages like Java, C++ and C, and is also acceptable from the performance point of view.

Python is a high-level, interpreted, interactive and object-oriented programming language. It was developed under an Open Source Initiative (OSI) approved open source licence. It is freely available, usable and distributable.

Python is designed to be easily readable. It has less syntactical construction than other programming languages, and is like the English language. The advantages of Python are:

- It's an easy and simple language to learn for beginners
- Allows rapid prototyping
- Syntax is easily readable
- There is a vast variety and number of libraries available, both inbuilt and community created
- Wide community support
- Easy to integrate

Python supports developers during the entire software development lifecycle, and helps them to be productive as well as confident about the product they are building. It is used for a wide variety of applications including creating a functional website, easy data analysis, and better integration with any environment, as it is a platform-agnostic language.

Characteristics of Python

Open source Python has the following characteristics.

- **Interpreted:** Processed at runtime by the interpreter. No need for compiling the program before executing it.
- **Interactive:** Can run in an interactive mode, i.e., the program can be written in a command line shell, which gives an immediate output for each statement.
- **Cross-platform:** Works on different platforms like Windows, Linux, Mac, etc.
- **Easy to learn** and easy to use, apart from being developer friendly.
- **Expressive, easy to read and understand:** The syntax of Python is like English statements.
- **Libraries:** Provides a large set of libraries, modules and functions that enable the rapid development of applications.
- **Integrated:** Easily integrated with other languages like Java, C/C++, etc.
- **Databases:** Provides interfaces for a large set of databases.
- **Object-oriented:** Supports the object-oriented style of programming that encapsulates code within the object.
- **Data types:** Has a variety of basic data types like integers, floating point number, strings (both in ASCII and Unicode), lists, dictionaries, etc.
- **Modules and packages:** Code can be grouped into packages and modules.
- **Memory management:** Automatic memory management deallocates the memory instead of manually handling it in the code.
- **Web applications** can be developed using Python. It provides libraries to handle protocols like HTML, XML, JSON, requests, etc. It provides various frameworks such as Django, Pyramid, etc, for Web application development.
- **Numeric computing applications:** Provides various libraries and packages like SciPy, Pandas, IPython, etc, for developing numeric and scientific computations.

Key open source packages in Python for machine learning

There are many open source packages in Python for ML problems. The ML solution is done based on the nature of the problem. Some of the most popular open source packages in Python for ML are listed below.

Numpy: This library provides support for large, multi-dimensional arrays and matrices, along with a large collection of high-level mathematical functions to operate on these arrays.

It handles linear algebra, Fourier transforms and random numbers. It interoperates with other libraries like Matplotlib, SciPy, Scikit-learn and TensorFlow. It is widely used in handling sound waves, images and other binary functions

Scikit-learn: This library provides a broad range of ML functions like classification, regression, clustering, dimensionality reduction, model selection, and preprocessing.

The Scikit-learn developer team maintains a strong focus on code quality and comprehensive documentation with the help of the open source community. It can interoperate with numeric and scientific libraries of Python like NumPy and SciPy.

Theano: This scientific computing library allows defining, optimising as well as evaluating mathematical expressions, which deal with multi-dimensional arrays. With Theano, making data-intensive calculations is up to a hundred times faster than when executing on a CPU alone.

Well optimised for GPUs and offering effective symbolic differentiation, Theano includes extensive code-testing capabilities. It can automatically avoid errors and bugs when dealing with logarithmic and exponential functions. Theano has built-in tools for unit testing and validation.

TensorFlow: This has an end-to-end library for performing high-end numerical computations. It can handle deep neural networks for